PCB Inspection Report

Timestamp	2026-04-04 16:26:37
Status	DEFECTIVE
Risk Level	HIGH
Failure Probability	70.06%

Detected Defects

#	Label	Confidence	BBox
1	Missing_hole	0.7954	[1192, 280, 1214, 301]
2	Missing_hole	0.6899	[802, 55, 848, 100]
3	Missing_hole	0.6693	[705, 55, 745, 98]
4	Missing_hole	0.649	[532, 274, 553, 297]
5	Missing_hole	0.5945	[108, 198, 131, 222]
6	Missing_hole	0.3384	[598, 56, 639, 98]
7	Missing_hole	0.3038	[954, 704, 973, 722]

Explanation

1) Root Cause Analysis
The consistent detection of seven distinct Missing_hole defects across the PCB, with high confidence levels ranging from 0.7954 to 0.3038, strongly indicates a systemic issue within the PCB fabrication process rather than isolated incidents. The most probable root cause lies upstream of final inspection. This could stem from errors in the Computer-Aided Manufacturing (CAM) data translation from design files, leading to an incorrect drill program. Alternatively, it points to significant drill equipment malfunction, such as sustained drill bit breakage or excessive wear not properly detected by tool monitoring systems, or consistent misalignment issues during the drilling operation itself. Less likely, but still possible, are issues with the original Gerber data lacking these specific hole definitions, though the presence of other features suggests a design intent. The distribution and number of these omissions suggest a global process deviation rather than a localized defect such as a single blocked hole after plating.
2) Operational Risks
The presence of multiple Missing_hole defects introduces severe operational risks, justifying the HIGH risk level. If these holes are intended as vias for critical signal routing or power/ground planes, their absence will result in immediate electrical open circuits, leading to complete functional failure of the board. Should these be component mounting holes, the mechanical integrity of mounted components will be compromised, leading to vibration-induced stress, component detachment, or improper seating. If thermal vias are missing, localized hotspots will develop, accelerating material degradation and device failure. The defect with 0.7954 confidence at [1192, 280, 1214, 301] and others with high confidence represent highly certain omissions that directly translate to these functional and mechanical failures, making a successful operational lifecycle improbable.
3) Failure Prediction

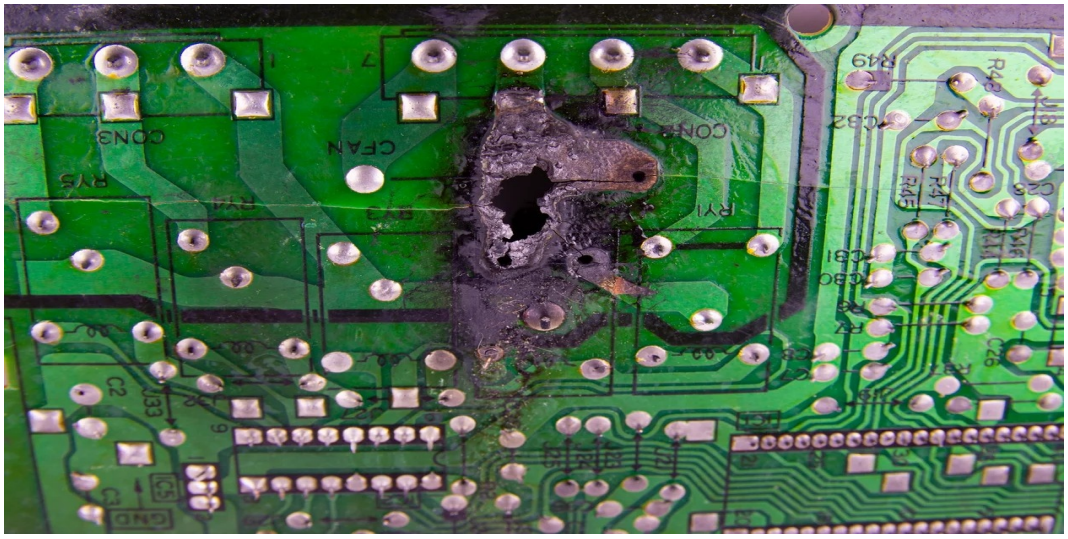
Given the 70.06% failure probability, the board is highly likely to fail upon initial power-up or during the first operational cycle. The most direct failure prediction is an immediate functional electrical failure due to uncompleted circuits if the missing holes are critical vias or component pads necessary for electrical connectivity. For instance, the high-confidence defects (e.7954, 0.6899, 0.6693) suggest critical omissions that would prevent the board from performing its intended electronic functions. If the missing holes are for mechanical mounting, failure will manifest as premature mechanical fatigue, component detachment, or structural instability under environmental stress (vibration, thermal cycling) during its operational life. The cumulative effect of multiple missing holes exacerbates these failure modes, ensuring poor reliability and early-life failures.
4) Repair Explanation
Repairing Missing_hole defects on a bare PCB is generally not feasible or recommended for high-reliability applications. A true "missing hole," especially if it is a plated through-hole or a blind/buried via, cannot be retroactively added without significant risk of damaging adjacent traces, layers, and dielectric material integrity. Attempting to drill and plate missing holes post-fabrication introduces micro-fractures, delamination risks, and compromises the precisely controlled impedance and layer registration of the original design. Such repairs would inevitably introduce new failure mechanisms and dramatically reduce the board's reliability. Therefore, for a board with multiple confirmed missing holes, particularly those with high confidence, the appropriate disposition is scrap due to the unrepairable nature of the fundamental structural and electrical defects.
5) Engineering Insights
To mitigate recurrence of these Missing_hole defects, several engineering controls are necessary. Firstly, implement a rigorous verification process for CAM data and drill programs against the original design files (Gerbers, ODB++). This should include automated Design Rule Checking (DRC) specific to hole presence and location. Secondly, enhance in-process monitoring and statistical process control (SPC) for drilling equipment, focusing on drill bit wear, breakage detection thresholds, and machine alignment calibration. Pre-drilling inspection of drill files and post-drilling inspection of panels (before plating) can catch these errors earlier in the fabrication flow, preventing further value-add to defective material. The AOI system, while effective in detecting these issues post-fabrication, should be calibrated for optimal detection of minimal hole diameters to ensure consistent identification, but the primary focus must be on preventing the defect at its source in the drilling stage.

Repair Suggestions

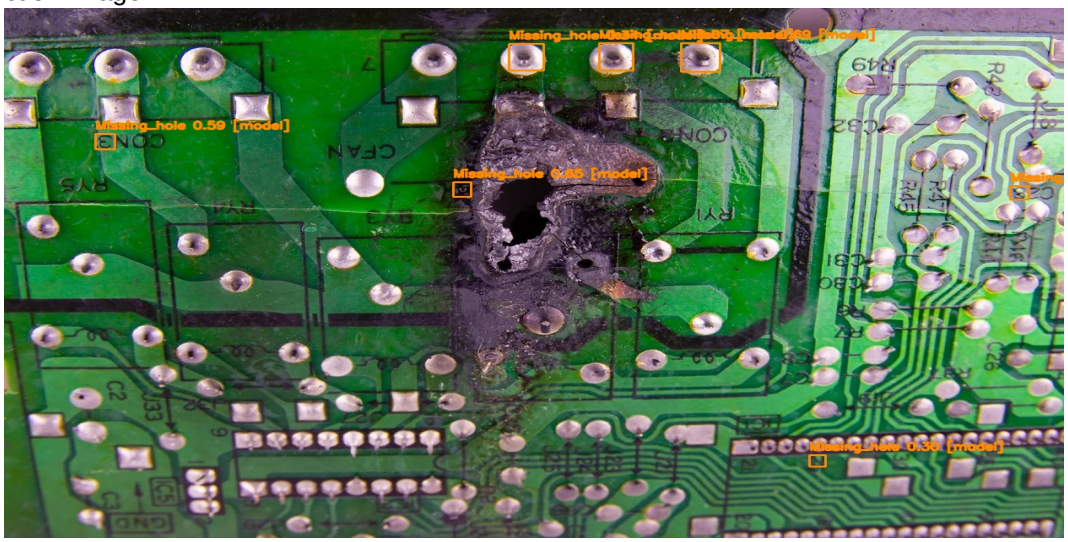
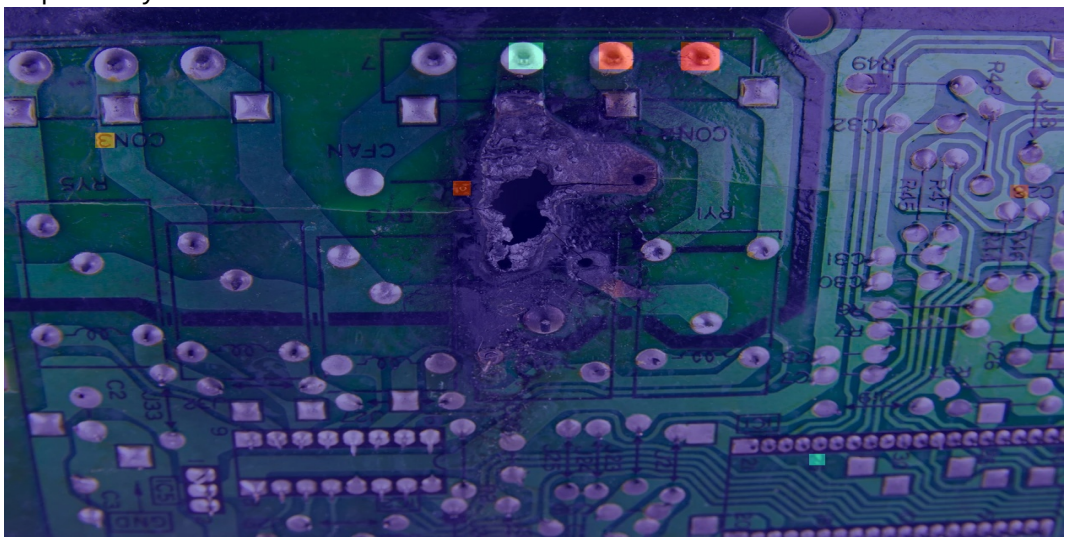
- Missing_hole: Replace the missing component and verify orientation and BOM value before reflow.

Inspection Images

Original Image



Detection Image



Heatmap Overlay